Lab 3: Application Requirements, HTTP and DNS

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3. Inspect a successful HTTP request

Record	Your observation
URL / request method	`http://127.0.0.1:8000/index.html` / `GET`
Status / Content-Type	`200 OK` / `text/html`
Response content / HTTP version	Page titled "Local networking lab", same as `web/index.html`. HTTP version: not displayed

4. Compare three service outcomes

- **A. Missing resource:** `http://127.0.0.1:8000/missing.txt` returned `404 File not found`.
- **B. Stopped server:** after Ctrl+C, `http://127.0.0.1:8000/index.html?test=stopped` showed "This site can't be reached, localhost refused to connect" (`ERR_CONNECTION_REFUSED`). No HTTP status.
- **C. Restored:** re-ran the same server command, `...?test=restored` returned `200 OK` with the page.

Case	Actual status or error	HTTP response received?
Working page	`200 OK`	Yes
Missing resource	`404 File not found`	Yes
Stopped server	`ERR_CONNECTION_REFUSED`	No
Restored service	`200 OK`	Yes

Differences

- **Working:** server, port and file all fine (GET `/index.html` gave 200).
- **Missing:** server answered with 404, so only that file is absent.
- **Stopped:** no response, nothing listening on port 8000, so the failure is the service not running.
- **Restored:** same command brought back 200, so the server process was the only cause.

5. Separate DNS from application access

DNS / application observation	Your result
Resolver / returned answer	Resolver `one.one.one.one` (`1.1.1.1`). Non-authoritative answer: `moodle.ccun.edu.kh` → `103.142.5.62`. Resolver IP = DNS server asked; returned IP = web server.

DNS / application observation	Your result
Public page outcome	`https://moodle.ccun.edu.kh/` loaded logged out: CCUN banner, Log in button, university tiles. No credentials entered.
What each test establishes	`nslookup` shows only that the name resolves. The browser shows that connectivity, TLS and the web app also work.

Why 127.0.0.1 works without DNS: it is already an IP address, so there is no name to look up.

Reflection

1. **Missing resource vs DNS failure:** a missing resource still gets an HTTP response (404). A DNS failure happens before any connection, so there is no HTTP response at all.
2. **Dev tools vs capture:** the browser is the endpoint and decrypts HTTPS. A capture on the wire only sees encrypted TLS data.
3. **Only one dependency changed:** 4B (stopped server). Restored by re-running `py -m http.server 8000 --bind 127.0.0.1 --directory web`.

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6. Requirements and acceptance tests

Numeric targets are proposed assumptions, not approved guarantees. Users and files are fictional.

Requirements

ID	Application	User	Location	Task	Proposed target	Dependencies
R1	Moodle	Student A	Student lab PC	Log in and download `week3-notes.pdf`	Page in ≤ 5 s, 2 MB file in ≤ 30 s, 30 students at once	DNS, Internet link, TLS, Moodle app, login
R2	Video meeting	10 participants	Classroom Wi-Fi	Join and talk with audio/video for 30 min	Join in ≤ 30 s, no audio drop > 2 s, ~720p video	DNS, Wi-Fi, bandwidth, meeting service, firewall
R3	Local printing	Staff B (authorized)	Office, same building network	Print 5-page `test-print.pdf`	Printed in ≤ 60 s, unauthorized users refused	Printer name/address, switch/VLAN, print server, authorization

Acceptance tests

ID	Req	Location	Action	Expected result	Evidence	Status
T1	R1	Student lab PC	Open Moodle, log in with test account, download `week3-notes.pdf`	File downloads fully, size/hash matches	Screenshot, download time, hash	Proposed
T2	R2	Classroom, 10 devices	Start test meeting, all join, 3 speak, 2 share video for 10 min	Join ≤ 30 s, clear speech, no freeze > 2 s	Join times, meeting stats, checklist	Proposed
T3	R3	Authorized PC, then unauthorized account	Print `test-print.pdf` from each	Authorized: 5 pages in ≤ 60 s. Unauthorized: refused	Printout, queue log, error message	Proposed

Failure scenario (one dependency)

DNS stops working for the student lab while everything else is fine. T1 fails ("server not found") and `nslookup moodle.ccun.edu.kh` errors out. Pinging the server's IP would still work, so ping alone would miss it.